



COURSE SYLLABUS

CSC16004 – Computer Vision

1. GENERAL INFORMATION

Course name: Computer Vision

Course name (in Vietnamese): Thị Giác Máy Tính

Course ID: CTT451

Knowledge block:

Number of credits: 4

Credit hours for theory: 45

Credit hours for practice: 30

Credit hours for self-study: 90

Prerequisite:

Prior-course:

Instructors: Tran Thai Son, Ph.D

2. COURSE DESCRIPTION

The computer vision course will provide knowledge about how to define and detect the features of images and video. Some basic problems of computer vision will be explained in this course such as image features, object detection, object tracking, object recognition, etc. This course will also provide skills in computer programming using OpenCV through its project final. It helps students implement knowledge of software engineering and computer vision into software application systematically.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
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G1	Understand knowledge about features of images and video	LO1
G2	Describe and understand about applications of computer vision	LO1
G3	Describe and understand basic solutions for fundamental problems of computer vision	LO1
G4	Describe and understand similarity measurement in Computer Vision	LO2
G5	Practice and enhance computer programming using OpenCV toolkit	LO2, LO3
G6	Attend classes to read, discuss and explain research papers or technical report	LO10, LO12

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Understand camera system	I/T
G1.2	Understand edge and contour features of images	I/T
G2	Know applications of computer vision	I/T
G3.1	Understand and explain local feature, global feature, and generic feature extraction in CV	I/T
G3.2	Understand and explain detection problem and its solution	I/T
G3.3	Understand and explain tracking problem and its solution	I/T
G4	Understand and apply distance measurement in CV	I/T
G5.1	Know and implement how to use OpenCV for implementing computer program.	T/U

G5.2	Know and implement the process of Computer vision to solve one specific problem	T/U
G6	Attend classes to read and understand CV methods through published papers in conference or journal of CV	U

5. TEACHING PLAN

THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Camera Modeling and Camera Calibration	G1.1	Deliver a lecture / Brainstorm / Discuss	Quiz
2	Image features, Edge, Measurement	G1.2	Deliver a lecture / Brainstorm / Discuss	Quiz
3	Project's introduction	G2	Deliver a lecture / Brainstorm / Discuss	Essay, Report
4	Corner detection, blob and DoG	G3.1	Deliver a lecture / Brainstorm / Discuss	Quiz
5	SIFT, HOG features	G3.1	Deliver a lecture / Brainstorm / Discuss	Quiz
6	Color and global features	G3.1	Deliver a lecture / Brainstorm / Discuss	Quiz
7	Generic features using deep learning model	G3.1	Deliver a lecture / Brainstorm / Discuss	Quiz
8	Similarity measurement	G4	Deliver a lecture / Brainstorm / Discuss	Quiz
9	Object detection & tracking	G3.2, G5.1, G6	Deliver a lecture / Brainstorm / Discuss	Essay
10	Open Problem & Discussion	G3.3, G5.2, G6	Deliver a lecture / Brainstorm / Discuss	Essay

LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Introduction of OpenCV	G5.1, G1.1	Deliver a lecture / Brainstorm / Discuss	
2	Image features, Edge, Measurement	G1.2	Deliver a lecture / Brainstorm / Discuss	Report
3	Project 's hint	G2	Brainstorm / Discuss	Report
4	Corner detection, blob and DoG	G3.1	Brainstorm / Discuss	Programming
5	SIFT, HOG features	G3.1	Brainstorm / Discuss	Programming
6	Color and global features	G3.1	Brainstorm / Discuss	Programming
7	Generic features using deep learning model	G3.1	Brainstorm / Discuss	Report
8	Similarity measurement	G4	Brainstorm / Discuss	Programming
9	Object detection	G3.2, G5.1, G6	Brainstorm / Discuss	Report, programming
10	Object tracking	G3.3, G5.2, G6	Brainstorm / Discuss	Report, programming

6. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
A1	Assignments	Implement algorithms of fundamental methods for each problem in computer vision.	G1.1, G1.2, G2	10-20%
A2	Projects	Utilize OpenCV toolkit and develop own idea to solve problems of computer vision such that	G3.1, G3.2, G3.3, G4, G5.1, G5.2, G6	40-50%

		students can understand their proposed idea thoroughly. Students will compare their project results and the results of the other method.		
A3	Final Exams	Essay and Quiz	G1.1, G1.2, G2, G3.1, G3.2, G3.3, G4	40-50%
A4	Attendant and activity in classes	Attend and discuss in classes	G6	5-10%

7. RESOURCES

Textbooks

- Richard Szeliski, Computer Vision: Algorithms and Applications (Texts in Computer Science), Springer 2010.

Others

- Nikos Paragios , Yunmei Chen , Olivier D. Faugeras, Handbook of Mathematical Models in Computer Vision, Springer 2010
- Olivier Faugeras, Three-Dimensional Computer Vision (Artificial Intelligence), The MIT Press 1993.
- Published papers in Computer Vision conferences.

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss on the topics. However, individual work must be done and submitted on your own.

